STA510 · BAYESIAN STATISTICAL METHODS

Do ER status and positive nodes relate to 5-year death in the same way in SEER and METABRIC?

A Bayesian comparison of two breast cancer datasets

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Introduction

Why compare SEER and METABRIC

Two datasets

- **SEER** (US registry)
- **METABRIC** (UK/Canada cohort)

Era, country and treatment differ between them.

Established prognostic factors

- ER status
- Number of positive lymph nodes

Do these associations transfer between populations?

An open question

- An earlier unadjusted analysis
- Found differences
- But with different outcomes and unequal follow-up (up to 107 vs 355 months).

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Research question

Research question

Are the associations of ER status and positive nodes with death within 5 years the same in SEER and METABRIC, and by how much do they differ?

DIFFERENCE

δ = β_METABRIC − β_SEER

log-odds scale, one for ER and one for nodes

INTERPRETATION

exp(δ) = ratio of odds ratios

1 = same association in both datasets
2 = one odds ratio is twice the other, a large difference
judged together with its 95% interval

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Data

Variables

Variable	What it is
ER status	Tumour cells have estrogen receptors (yes/no)
Positive nodes	Lymph nodes with cancer cells (count)
Age	At diagnosis (years)
Tumour size	Size at diagnosis (mm)
Tumour grade	How abnormal the cells look (1–3)
Survival months	Follow-up time from diagnosis
Status	Alive or dead at last follow-up

Outcome and patient selection

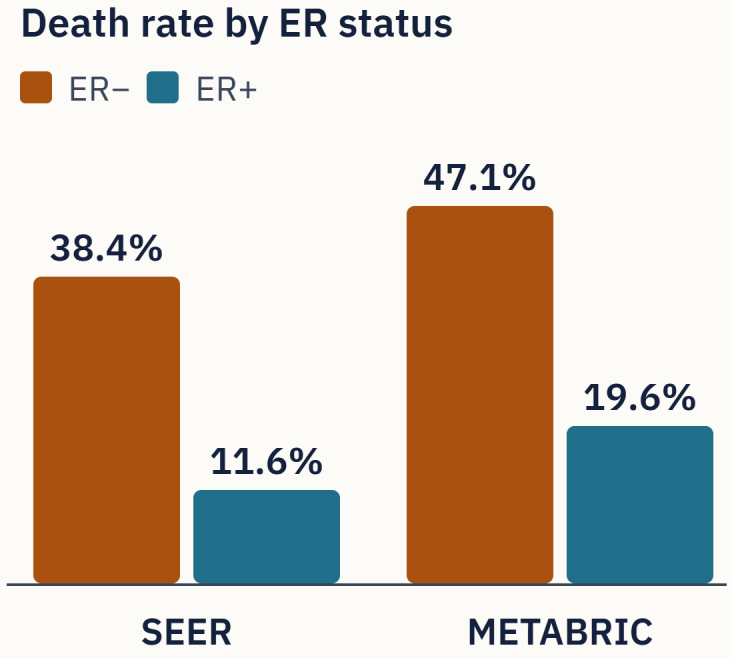
Outcome: death within 5 years (60 months); follow-up differs (107 vs 355 months).

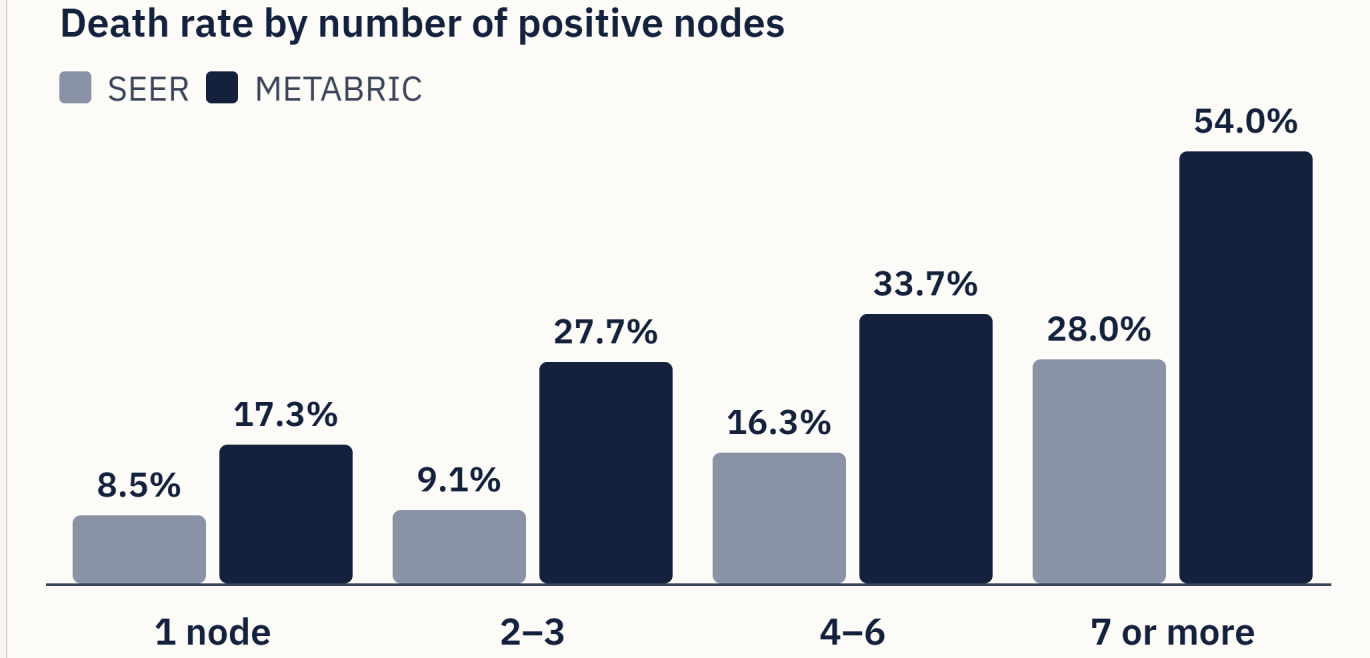
Patients	Raw	Removed	Analysed
SEER	4,024	−772	3,252
METABRIC	2,509	−1,933	576

Removing criteria:

- **No status or time:** the outcome cannot be built.
- **Alive with under 60 months of follow-up:** outcome unknown (deaths are kept).
- **Missing covariates:** the model needs all five.
- **Age 30–69, ≥1 node (METABRIC):** makes it resemble SEER.

Analysis sample





3,252 SEER and **576** METABRIC patients; outcome: death within 5 years. Crude rates, not adjusted: ER− and more positive nodes go with higher death rates in both datasets.

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Methods

The model, its priors and the four models compared

Model (M1, the main model)

$$y_i \sim \text{Bernoulli}(\mu_i)$$

$$\text{logit}(\mu_i) = \alpha_{j(i)} + \beta_{1,j(i)} X_1 + \beta_{2,j(i)} X_2 + \beta_3 X_3 + \beta_4 X_4 + \beta_5 X_5$$

$y_i = 1$ if patient i died within 5 years; $j(i)$ = dataset of patient i

Separate per dataset

α intercept
β_1 ER+ vs ER−
β_2 log(1 + nodes)

Shared

β_3 age (per year)
β_4 tumour size (per mm)
β_5 grade (per step)

Dataset difference

$\delta = \beta_{METABRIC} - \beta_{SEER}$, computed from the posterior draws of β_1 and of β_2

Priors

Parameter	Prior N(mean, sd)	Source
α · intercept	N(−1.815, 1)	NCI, Survival Rates and Prognosis for Breast Cancer (SEER 2016–2022, regional stage)
β_1 · ER+ vs ER−	N(−0.57, 0.5)	Bentzon et al. 2008, Int J Cancer 122:1089–94; Moffat 2014, Diseases of the Breast, ch. 28
β_2 · log(1 + nodes)	N(0.7, 0.4)	Moffat 2014, ch. 28, fig. 28-3 (SEER data)
β_3 · age	N(0.0203, 0.0101)	Kim et al. 2022, The Breast 61:77–83; Dai et al. 2019, PeerJ 7:e7252
β_4 · size	N(0.0265, 0.0133)	Moffat 2014, ch. 28, fig. 28-3 (SEER data)
β_5 · grade	N(0.4, 0.25)	BMC Cancer 2006;6:194

Centre and sd: published estimate → log odds ratio per unit → shrunk (unadjusted source, other horizon) → range as centre ± sd. The same priors are used for SEER and METABRIC.

Models compared

Model	ER and nodes	Age, size, grade	Pooling
M0	one shared coefficient ($\delta = 0$)	shared	complete
MH	$\beta_j = \beta_0 + u_j$, $u_j \sim N(0, \sigma^2)$, $\sigma \sim$ exponential(2)	shared	partial
M1 (main)	separate per dataset	shared	none
M2	separate per dataset	separate per dataset	none

α is separate per dataset in all models. In MH, β_0 is the common mean and u_j the deviation of dataset j; σ is a standard deviation.

Fitting and checking

Fit

- Hamiltonian Monte Carlo (Stan via brms)
- 4 chains × 2,000 iterations, half of them warm-up: 4,000 draws
- MH: adapt_delta 0.99 after 9 divergences at 0.95

Convergence

- $\hat{R}$ = 1.00 (rule: below 1.1)
- ESS above 2,800 for M1 and M2; lowest 1,050 for MH
- No divergences, clean trace plots

Model checks

- Prior predictive: do the priors imply plausible death rates?
- Posterior predictive: does the model reproduce the observed rates by group?
- loo: which model predicts new patients best?

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Results

Differences between datasets, model checks and comparison

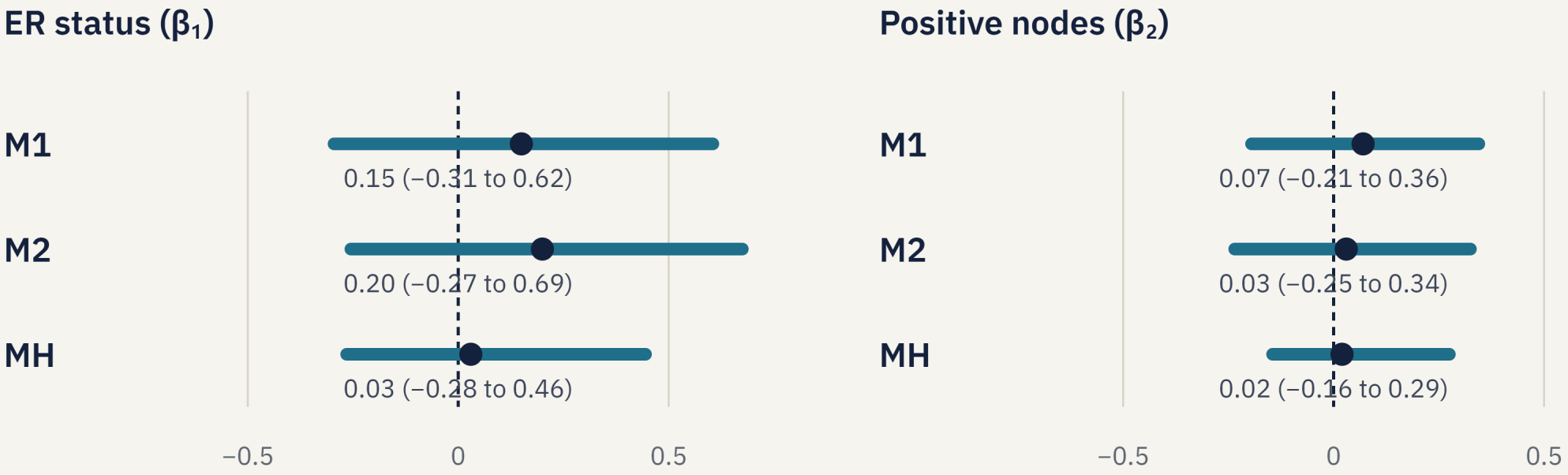
Prior predictive check

Implied 5-year death rate	Median	95% range
SEER	24.0%	4.9–62.6%
METABRIC	27.6%	6.2–66.3%

- Simulated from the priors alone, over all patients (4,000 draws)
- Wider than a real cohort, but plausible across patient types, so the priors were kept

Dataset difference δ

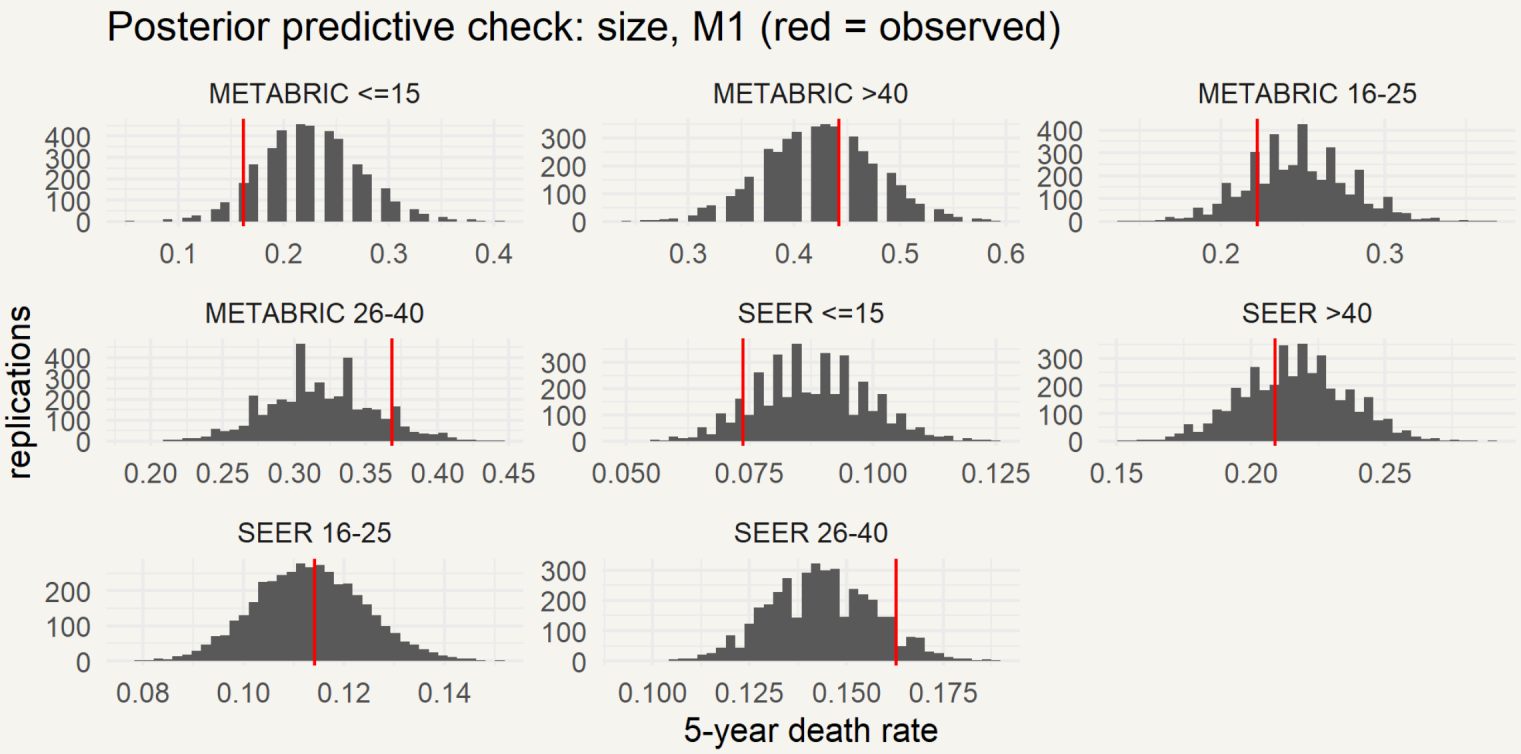
How to read it: exp(δ) = 1 means the same in both datasets. Above 1: weaker ER protection, stronger node effect in METABRIC.



δ on the log-odds scale: dot = posterior median, bar = 95% interval

All six 95% intervals include 0, and for ER a ratio of odds ratios up to about 2 is not excluded.

Posterior predictive check: size



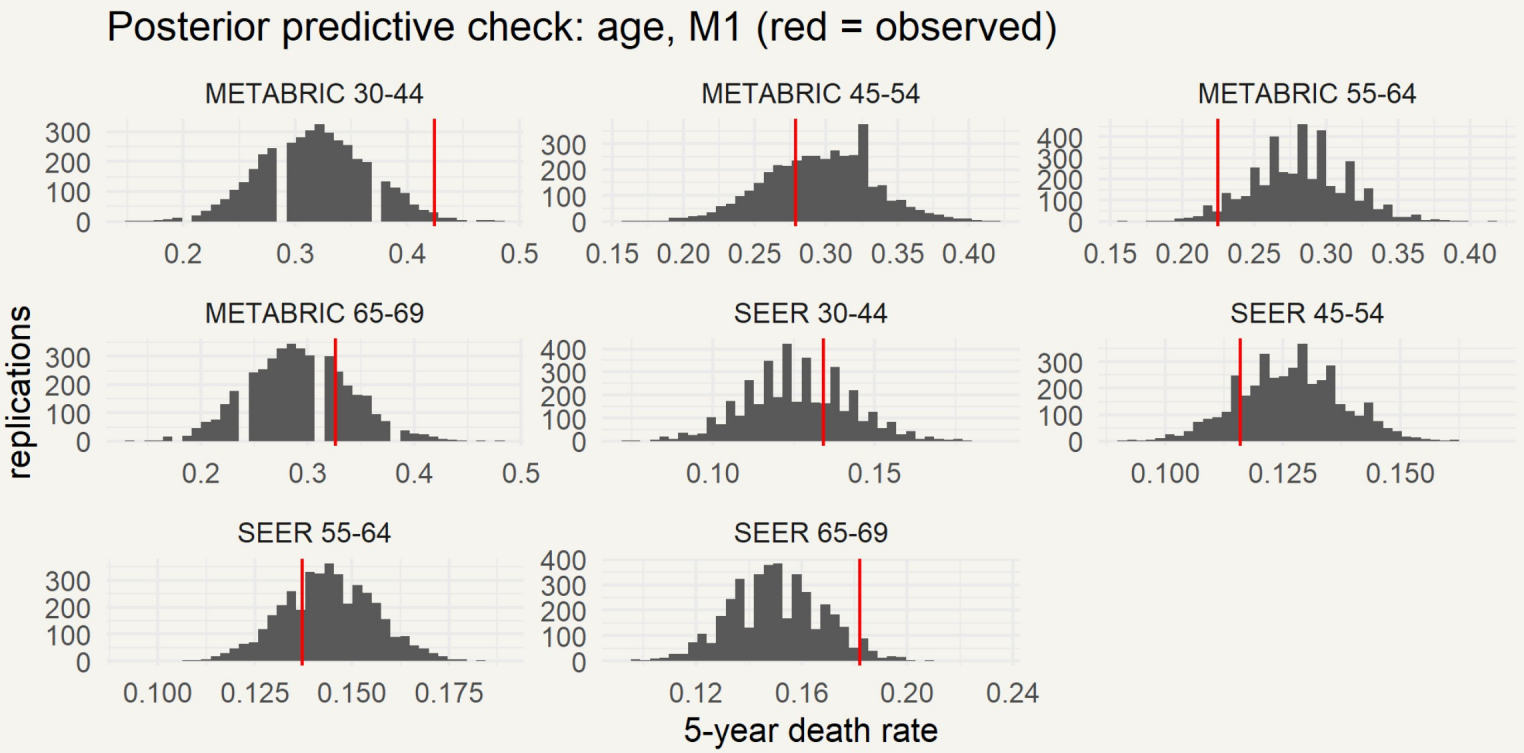
How to read it

Bars: replicated death rates from the posterior. Red line: observed rate. A failure is a red line in the tail.

What we see

- ≤15 mm: observed below the replications
- 26–40 mm: observed above them
- Same pattern in both datasets: a non-linear size effect
- ER, nodes and grade: no misfit (backup)

Posterior predictive check: age



What we see

- METABRIC 30–44: above (p 0.021)
- METABRIC 55–64: below (0.972)
- SEER 65–69: above (0.046)
- Same pattern under M2: a shape problem (U-shape), not a dataset-specific slope

The linear age term is an approximation.

Model comparison

Pooling of ER and nodes	elpd_diff (se)
M0 complete	0
MH partial	−0.9 (0.4)
M1 none	−1.7 (0.8)

Age, size, grade	elpd_diff (se)
M1 shared	0
M2 separate	−0.2 (1.8)

- All |elpd_diff| are below 4: the models predict about equally well
- Pareto k below 0.7 for all four models, so the leave-one-out approximation is reliable
- Weak support for pooling, not evidence that the relationships are identical

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Discussion

Discussion

Interpretation

- No clear evidence of a difference in the ER or the node association
- Not evidence of sameness: for ER a ratio of odds ratios up to about 2 is compatible with the data
- The earlier differences may come from unequal follow-up and no adjustment (not tested)

Limitations

- Excluding patients with an unknown 5-year outcome assumes censoring is unrelated to the outcome
- Linear age and size terms fit only approximately
- With two groups, the data say little about how much to pool

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Conclusions

The answer and next steps

Conclusions

No clear evidence that the ER or node association with 5-year death differs between SEER and METABRIC; a moderate difference is not excluded.

NEXT STEPS

- Log tumour size and a flexible age term
- Prior sensitivity analysis
- Repeat with the unrestricted METABRIC sample

Thank you

Backup: MCMC and convergence

What MCMC does

- Posterior ∝ prior × likelihood, with no closed form for nine parameters
- HMC/NUTS draws samples that follow the posterior density
- 4 chains × 1,000 post-warm-up draws; every quantity (δ, exp(δ), P(δ > 0)) is computed per draw

Check	Criterion	Result
Trace plots	chains overlap	clean
$\hat{R}$	< 1.1	1.00
Bulk ESS	≥ ~1,000	M1 > 2,800; MH ≥ 1,050
Divergences	0	MH: 9 at adapt_delta 0.95, 0 at 0.99

Convergence says the sampler works for this model. Whether the model fits is checked separately, with posterior predictive checks and loo.

Backup: δ in numbers

Effect	Model	Median δ	95% interval	exp(δ) (95%)	P(δ > 0)
ER	M1	0.15	−0.31 to 0.62	1.16 (0.73 to 1.86)	0.74
ER	M2	0.20	−0.27 to 0.69	1.23 (0.76 to 1.99)	0.80
ER	MH	0.03	−0.28 to 0.46	1.03 (0.76 to 1.59)	0.62
Nodes	M1	0.07	−0.21 to 0.36	1.08 (0.81 to 1.43)	0.70
Nodes	M2	0.03	−0.25 to 0.34	1.03 (0.78 to 1.40)	0.59
Nodes	MH	0.02	−0.16 to 0.29	1.02 (0.85 to 1.34)	0.64

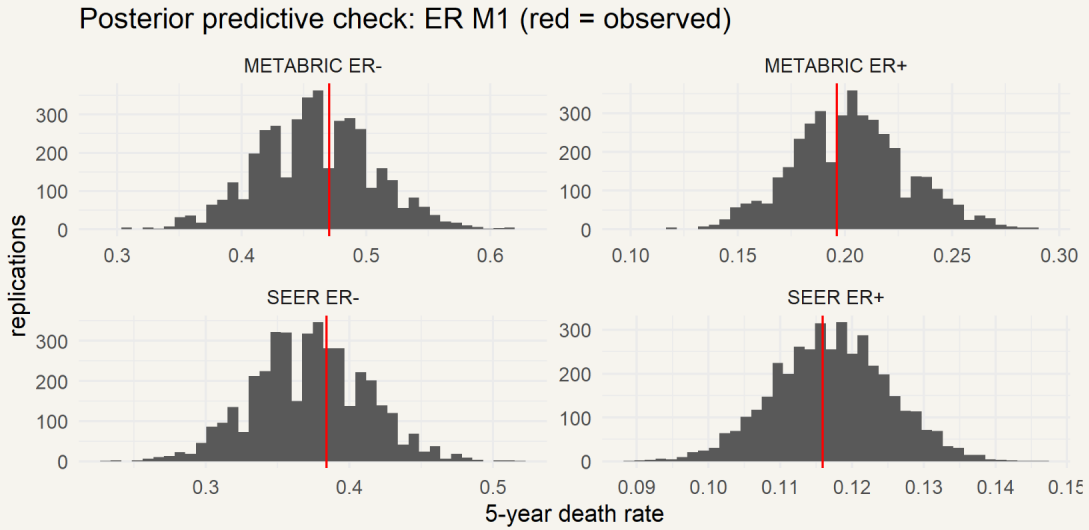
In MH the posterior median of σ is 0.18 (ER) and 0.13 (nodes), but its 95% upper limits are 1.12 and 0.78. With two groups the data say little about σ, so the narrower MH intervals partly reflect the assumption that the datasets are similar.

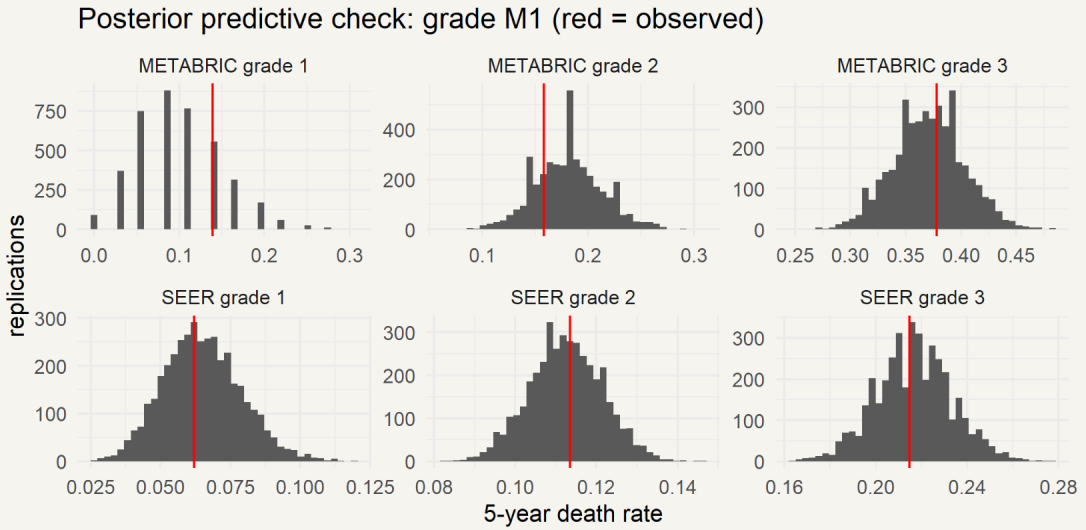
Backup: Table 1

Before restriction	SEER	METABRIC
Patients	3,252	1,760
Died within 5 years	442 (13.6%)	398 (22.6%)
ER-positive share	92.6%	76.3%
Death rate, ER+	11.6%	17.6%
Death rate, ER−	38.4%	38.8%
Age, median (quartiles)	54 (47–62)	61.7 (51.3–70.6)
Positive nodes, median (quartiles)	2 (1–5)	0 (0–2)
Tumour size mm, median (quartiles)	25 (16–39)	23 (17–30)

Analysis sample: METABRIC restricted to age 30–69 and at least 1 positive node, like SEER: 576 patients (death rate ER− 47.1%, ER+ 19.6%).

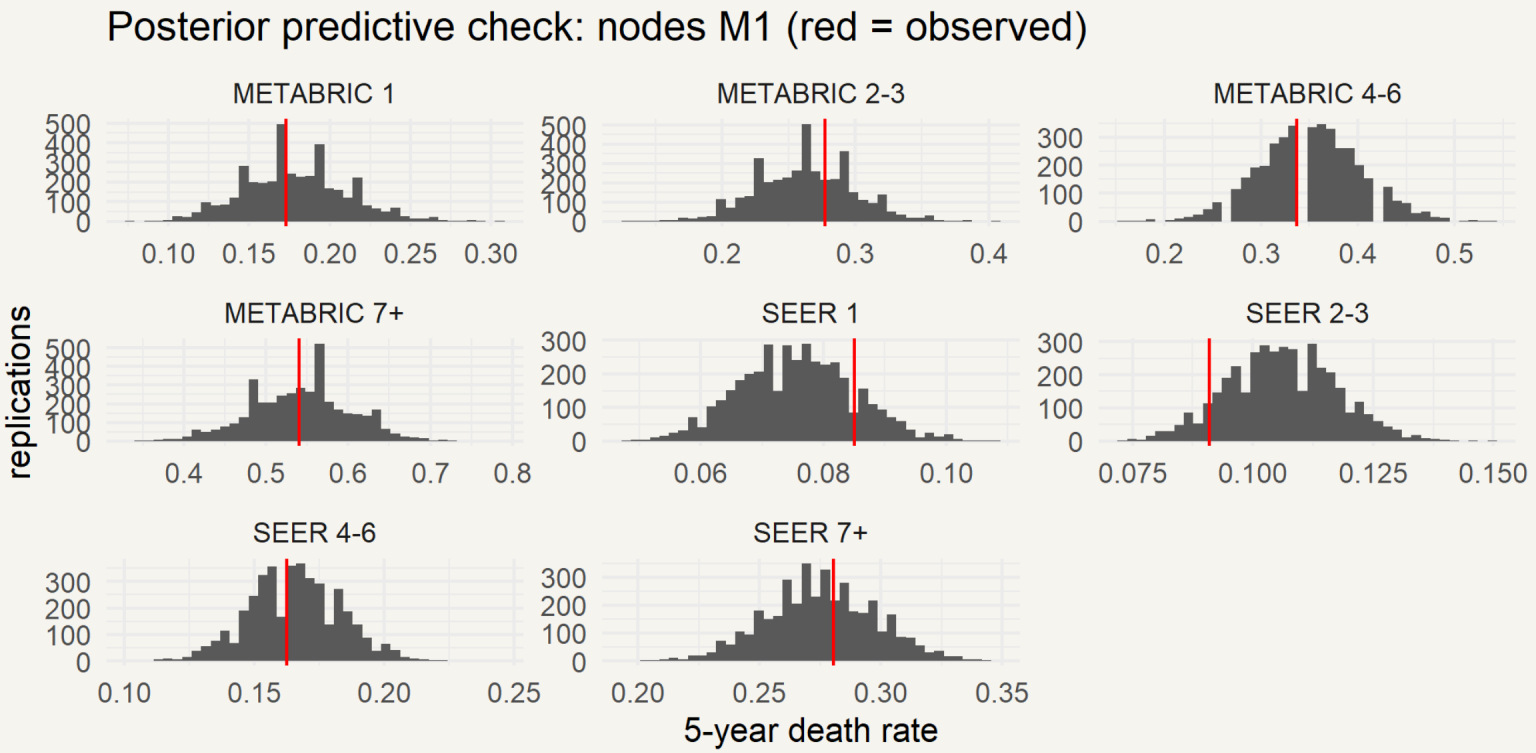
Backup: posterior predictive checks, ER and grade





Model M1. Bars: replicated death rates; red line: observed rate. In every panel the observed rate lies inside the replications, so ER and grade show no misfit in either dataset.

Backup: posterior predictive check, positive nodes



What we see

- Observed rates inside the replications in all eight bands
- SEER 2–3 nodes is the most extreme (p = 0.92, just beyond the 0.89 rule of thumb); SEER 1 node p = 0.17
- In SEER the rate is flat from 1 to 2–3 nodes, flatter than the model: a mild shape issue, as for size and age